PROJECT REPORT

Supply Chain Intelligence

Delivery, Inventory & Profit Analysis

An End-to-End Data Analysis Project

With Python | Excel | Power BI

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Objective

The objective of this project is to analyze supply chain performance by identifying inefficiencies in delivery, inventory management, and profitability. The goal is to generate actionable insights to improve operational efficiency and business outcomes.

Tools

Python, Excel, Power BI , Canva for background designing

Data Overview

- Source (Kaggle)

The dataset consists of approximately 180,000 records containing order-level supply chain information, including sales, delivery timelines, customer segments, and product details. This data enables analysis of logistics efficiency, inventory management, and profitability.

Methodology

Data Cleaning

- Python (Pandas)
- Excel (manual cleaning, duplicates removal)

Feature Engineering

- Delivery gap
- Stockout flag
- High-value orders

Visualization

- Power BI dashboard

Design

- Canva used for dashboard background

The approach focus on transforming raw data into actionable insights through structured analysis and visualization.

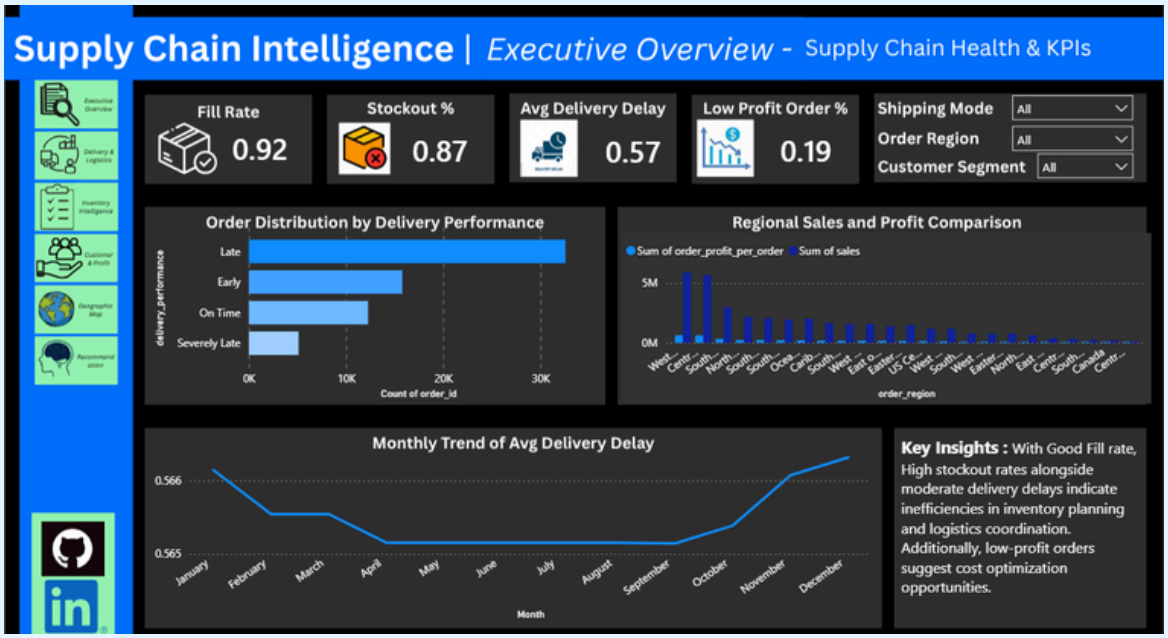
Key KPIs

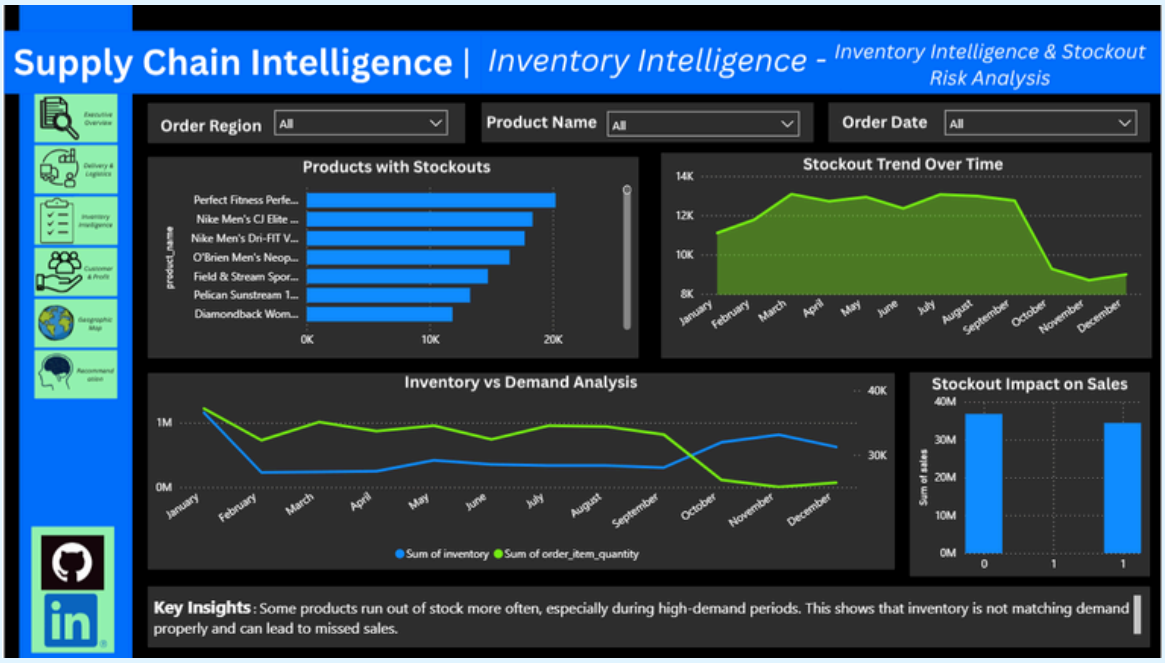
- Fill Rate → Measures demand fulfillment efficiency
- Stockout % → Indicates inventory shortages
- Avg Delivery Delay → Measures logistics performance
- Low Profit Orders % → Identifies margin inefficiencies

Key Insights

- A high stockout percentage indicates inefficiencies in inventory planning and demand forecasting.
- Delivery delays are higher in standard shipping, indicating inefficiencies in cost-optimized logistics.
- Revenue is driven by regular orders rather than high-value orders.
- Some regions show consistent logistics inefficiencies.

Dashboard Overview





Power BI Dashboard Overview and supply chain insights.

Recommendation

- **Delivery optimization** : Improve route planning and optimize standard shipping to reduce delays.
- **Inventory planning** : Implement demand forecasting to align inventory with sales trends, always maintain back-up stock of high-demand products.
- **Profit improvement :** Focus on reducing low-margin orders by optimizing pricing strategies and controlling shipping costs to improve overall profitability.
- **Customer strategy :** Prioritize high-value customers by offering faster delivery and better service to improve retention and long-term revenue.
- **Regional improvement** : Strengthen logistics and inventory planning in high-demand regions to reduce delays and improve efficiency.

Challenges Faced

- Data inconsistency and duplicate columns
- Difficulty in handling large dataset
- Import issues during data loading
- Designing meaningful KPIs

Learning

- Improved data cleaning and feature engineering skills
- Learned how to convert business problems into KPIs
- Gained experience in dashboard design and storytelling
- Understood real-world supply chain challenges

Business Impact

This project demonstrates how data-driven insights can improve supply chain efficiency by reducing delivery delays, minimizing stockouts, and optimizing profitability through better operational planning.